Rigid-Body Symmetry in Flow-Matching Motion Planning: What Canonicalisation Buys, and What It Does Not

Andrea Sevincel   Xinzhe Zhou   Xiaoming Duan

Shanghai Jiao Tong University

aemirsevincel@gmail.com   zhou_xinzhe@sjtu.edu.cn   xduan@sjtu.edu.cn

Abstract

This is a controlled study of how to exploit rigid-body symmetry in a generative motion planner, not a proposal for a competitive planner. Motion planning is exactly equivariant under rigid transformations, and a substantial literature builds that symmetry into network architectures. We hold the architecture, data and training budget fixed, vary only the representation of the problem, and find that almost all of the available benefit is obtained without touching the architecture at all. We then measure how far the trained field departs from equivariance, prove exactly what that quantity bounds — the error any post-hoc symmetrisation of a given model can remove — and show what it does not bound: two arms that are equally equivariant differ by 28 points of held-out performance. Equivariance of the learned field and performance are separable, and the measurement sees only the first.

Our starting point is that the start and goal of a planning query themselves define a frame, and that this frame removes five of the six degrees of freedom of SE(3) exactly, at zero computational cost. The sixth — a roll about the start–goal axis — cannot be removed by any continuous rule; we give two independent obstructions, one topological and one distributional. We therefore treat the sixth by symmetry: roll augmentation during training and frame averaging at inference.

The size of the effect is best read against a floor rather than in isolation. On a cluttered 3D point-mass benchmark, world-frame flow matching reaches 15.4% of individual samples collision-free on held-out environments — statistically indistinguishable from the 15.6% obtained by joining start to goal with a straight line, measured on the identical problems. Per sample, training on raw world coordinates buys nothing over joining the endpoints with a line. With the reduction, the same network, data and budget reach 45.5%, and, contrary to the hypothesis that canonicalisation is a small-data crutch, the gap *widens* monotonically with training data: the world-frame arm gains 2.6 points across a 12.5× increase in training environments while the canonicalised arm gains 26.2 and has not plateaued. Every cell of that grid is a mean over three seeds, and the gap at 250 environments is $+30.1 \pm 0.4$ points across seeds.

All three mechanisms built for the sixth degree of freedom are worth almost nothing at convergence: seed-matched, frame averaging contributes +0.68 points, roll augmentation $+0.8 \pm 0.3$, and an exactly SO(2)-equivariant backbone +0.05 — three unrelated ways of imposing the same symmetry, all recovering nothing. The constrained architecture is not useless, however: it reaches in a quarter of the epochs what the unconstrained one needs to converge, so the symmetry buys optimisation rather than accuracy. A translation-only reduction splits it: recentring alone is worth +12.3 and aligning the axis a further +18.0. We explain this rather than merely reporting it, by measuring the *non-equivariance residual*: only 1.7% of the learned velocity field lies outside the equivariant subspace. Because frame averaging is the orthogonal projection onto that subspace, 1.7% is the entire budget any post-hoc symmetrisation of this model can spend. We propose the residual as a cheap diagnostic of how much non-equivariant structure a learned field retains: it is one forward pass, needs no retraining, and it quantifies the headroom available to post-hoc symmetrisation exactly. The separation asserted above is measured, not assumed: under the full SE(3) action, augmentation and canonicalisation leave the field equally equivariant ($r = 0.018$ against 0.017) while differing by 28 points.

The obvious alternative — exploiting the same symmetry by augmenting training with random rigid motions — recovers little of this: at 250 environments it lifts the world-frame arm from 15.4% to 17.5%, about 7% of what the reduction achieves. We also find that optimisation and representation are *separate* bottlenecks: tripling the training budget is worth +8.0 points to the canonicalised arm and +0.9 to the world-frame one, so compute cannot buy what the representation forecloses — and on the deployment metric the same tripling moves the world-frame arm *backwards*. The pattern replicates in a second state space, an L-shaped rigid body whose state is a full pose rather than a point: the world-frame arm fails to clear its own trivial floor in both domains, the reduction clears it in both, and the ratio between arms is 2.9× in each. Finally we bound our own contribution from above. Appending the signed distance to the scene and its gradient at each waypoint — a featurisation of obstacle data the network already receives, not privileged information — is worth +39.5 points against the world frame where the reduction is worth +35.8, so the representation change studied here is not the largest effect available on this benchmark. It is, however, not merely a substitute for that information: with local geometry supplied, the reduction is still worth +24.7 points. We calibrate both ends on the same 500 problems: an untrained Brownian-bridge prior, swept over widths, never exceeds 15.8% at best-of-20, while classical planners reach 74–100% in under 0.4 s on one CPU core. The learned sampler does not approach the latter, and we present this as a controlled study of representation choices, not as a planner proposal.

I Introduction

A motion planning problem consists of a workspace populated with obstacles, a start configuration $\mathbf{s}$, and a goal configuration $\mathbf{g}$; its solution is a path τ joining them and avoiding the obstacles. The problem carries an exact symmetry: for any rigid transformation $T \in \mathrm{SE}(3)$,

$$\tau\big(T\cdot\mathcal{O}, T\mathbf{s}, T\mathbf{g}\big) \;=\; T\cdot\tau\big(\mathcal{O}, \mathbf{s}, \mathbf{g}\big), \tag{1}$$

where $\mathcal{O}$ denotes the obstacle set. Nothing about the geometry changes when the problem is picked up and moved.

A neural planner trained on raw world coordinates has no access to Eq. (1). As far as its weights are concerned, planning left-to-right near the ceiling is an unrelated task from planning front-to-back near the floor, and the same skill must be relearned at every pose. The dominant response in the literature is architectural: constrain the layers so that equivariance holds by construction [5, 26, 9, 6, 10], an approach that has been carried into robot learning with real success [23, 28, 25, 29]. Our question is not whether that route works — it does — but how much of its benefit is available without it. We are deliberate about not asserting more: this paper does not benchmark a constrained backbone, so it is in no position to claim that constrained layers cost expressiveness or generalise worse, and we make no such claim. What we do establish is a lower bound on what the *unconstrained* route reaches, obtained from a frame that is read off the query in closed form.

There is a second route, older and much cheaper: build the symmetry into the *representation of the problem*, so that the network is never shown the pose and therefore cannot be sensitive to it [20, 13]. This paper asks how far that route goes on a generative motion planning problem, and what can — and cannot — be inferred from measuring how equivariant a trained model already is.

Our answer has five parts.

(i) Five of six degrees of freedom are free. The query supplies a frame. Placing the origin at the midpoint of $\mathbf{s}$ and $\mathbf{g}$ and aligning the first axis with the start–goal direction removes three translational and two rotational degrees of freedom, exactly and at no cost. The resulting representation is invariant under any rigid transformation of the problem *up to a single roll about the start–goal axis*. The quantities a roll leaves alone — the separation d, the first coordinate of every reduced point, and its radius in the plane normal to that axis — are invariant outright (Proposition 1). Within those five that is a stronger guarantee than a trained equivariant architecture provides, since it holds to floating-point precision at initialisation rather than to optimisation tolerance. The reduction therefore collapses SE(3) to SO(2); the remaining roll is what the rest of the paper is about. As a side effect the six numbers previously used to condition the network on the query collapse to a single invariant scalar.

(ii) The sixth is provably not removable, for two unrelated reasons. No continuous rule can fix the roll about the start–goal axis: such a rule is a nowhere-vanishing tangent field on S^2, which the hairy ball theorem forbids (Proposition 2). Deriving the roll from the scene instead also fails, because the obstacle distribution is symmetric under central inversion, so the first angular moment — the one a scene-derived gauge would actually use — vanishes in expectation for every start–goal direction (Proposition 3). We therefore handle the sixth degree of freedom by symmetry rather than canonicalisation.

(iii) The symmetry machinery buys little, and we measure why. Roll augmentation and frame averaging contribute +0.8 and +0.1 percentage points, neither distinguishable from zero across three seed-matched runs, while the reduction contributes +30.1±0.4. Rather than infer inertness from flat curves — a weak conclusion — we measure the quantity frame averaging exists to remove. The *non-equivariance residual*, the disagreement among rotated copies of the velocity field, is 1.7% and is flat across a 12.5× range of training data. Since frame averaging is the orthogonal projection onto the equivariant subspace (Proposition 4), the error it can remove is bounded by that 1.7% (Corollary 5), while the model is failing on most of its samples: the model's error is overwhelmingly equivariant error.

(iv) The third mechanism, and what it does buy. A symmetry can be imposed in the data, in the operator, or in the weights. Earlier versions of this work measured the first two and recorded the third as untested, declining to let Corollary 5 stand in for it. We now build an exactly SO(2)-equivariant backbone — verified to 1.2×10^{-6} on an *untrained* network, and carrying 19.5% *more* parameters than the unconstrained one, so the comparison is not confounded by capacity. It is worth +0.05 points at convergence and roughly a fourfold saving in optimisation. The caution was justified in both directions: the residual's smallness did not predict the first number, and nothing about r could have predicted the second.

(v) Calibration against floors and ceilings, including an upper bound on our own contribution. A collision-free rate is uninterpretable on a new benchmark, so we measure, on the identical 500 held-out problems, both a trivial lower bound and four classical planners. The trivial bound is the finding: a straight line from start to goal is collision-free on 15.6% of problems, and the world-frame model scores 15.4% per sample. The classical planners (74–100%, all under 0.4 s on a single CPU core) bound the other side and show that the learned sampler, at 45.5%, is not competitive with them here. Because a generative planner is deployed as best-of-N behind a collision check, we also report that per-query metric against the honest control for it — the model's own untrained prior, sampled N times, at its best width. That comparison separates two things the per-sample number conflates: the world-frame arm is at the straight-line floor per sample, but well above the untrained prior per query, so it has learned sample *diversity* without learning sample *quality*.

A reader may reasonably ask why a generative planner is worth studying on a problem with an exact SDF, where RRT-Connect solves essentially everything in a third of a second. We answer it here rather than by implication. PointMass3D is an analytical sandbox chosen *because* the SDF is exact: an analytic collision checker makes the floor and the ceiling computable on the identical problems, so a representation change can be measured against both rather than only against other configurations of itself. On this benchmark one should use RRT-Connect, and we say so in Sec. V-B. The measurement is worth making anyway because the quantity being isolated — what a generative planner gains from being shown its query in a canonical frame — is a property of the representation rather than of the collision checker, and it transfers to the settings where generative planners are the right instrument: learned or perceptual cost fields, contact-rich dynamics, multi-modal solution sets. None of those admit an analytic SDF, and a result of this kind can only be isolated where the confounds are removable.

We regard (iii) as the transferable contribution. The residual is a few lines of code, costs one forward pass, requires no retraining, and can be computed on any model with a group action on its inputs. It converts "we tried equivariance and it did not help" — an anecdote — into a quantitative statement about how much headroom *post-hoc symmetrisation* has on a given problem, with an exact bound attached. We recommend it as a diagnostic to run before reaching for symmetrisation, and we are explicit about its limit: measured across three arms of this benchmark, r does not predict what a symmetry mechanism is worth. Two models with indistinguishable residuals differ by 28 points (Table 5), because one of them fixed the representation and the other only learned the symmetry.

II Related Work

Equivariant architectures. Group-equivariant convolution [5] generalises translation weight-sharing to larger groups; for SO(3) and SE(3) the standard constructions are tensor field networks [26], SE(3)-Transformers [9], vector neurons [6], and the `e3nn` framework [10]; [2] surveys the area. In robot learning these ideas underpin neural descriptor fields [25], equivariant descriptor fields [23], equivariant diffusion policy [28], and SIM(3)-equivariant policies [29]. This work does not dispute those results; it provides a measurement that indicates when they are the right instrument.

Canonicalisation and frame averaging. Frame averaging [20] obtains equivariance by averaging an unconstrained network over a set of frames derived from the input, and learned canonicalisation [13] predicts a single frame with a small auxiliary network. Both leave the backbone unconstrained. Our $(\mathbf{s}, \mathbf{g})$ reduction is a canonicalisation in this sense, with the distinguishing feature that the frame is not learned or searched but read directly off the query, so five degrees of freedom are removed in closed form; the obstruction of Proposition 2 is precisely the reason the sixth cannot be handled the same way, and is why we fall back to frame averaging there.

Generative motion planning. Diffusion and flow models over whole trajectories have become a standard planning substrate [12, 4, 3, 27]. We use the conditional flow matching / rectified flow formulation [15, 16, 1] for its straight-line interpolants and few-step sampling. Learned planners that regress directly to paths [22, 8] form the non-generative counterpart. Within this planning literature we are not aware of a reported measurement of how far the learned field departs from equivariance.

Measuring learned equivariance. The idea of measuring, rather than assuming, how equivariant a trained network is, is not new, and the closest work to ours is outside planning. Gruver et al. [11] measure learned equivariance through the Lie derivative of a trained model with respect to the group action, and find that unconstrained architectures often acquire substantial equivariance from data alone — the same qualitative phenomenon our residual reports, in a different parameterisation and for continuous groups. Elesedy and Zaidi [7] analyse the generalisation benefit of equivariance using the orthogonal decomposition of a function into equivariant and anti-equivariant parts, which is the decomposition Corollary 5 is stated in. Relative to these, what we add is narrow and practical: the residual is a single forward pass with no differentiation and no retraining, and it comes with an exact statement of what it bounds. We are careful not to claim more: it measures the headroom for symmetrising a *given* model, which is not the same question as how much a constrained architecture would gain.

Classical planners supply our supervision: RRT-Connect [14] for feasibility, then shortcutting and CHOMP [30] refinement, with TrajOpt [24] available in the benchmark.

III Problem Setup

A Benchmark

Experiments use PointMass3D, a benchmark we built for this study. A spherical robot of radius 0.03 moves in the workspace $[-1, 1]^3$ populated with 20 spheres and 20 axis-aligned boxes per environment, obstacle centres drawn uniformly in $[-0.75, 0.75]^3$. Collision checking uses an analytic signed distance field: the environment SDF is the minimum over obstacle SDFs and the workspace walls, and clearance subtracts the robot radius. Start and goal are drawn uniformly subject to a collision margin and a minimum separation $\|\mathbf{g}-\mathbf{s}\| \geq 1.5$. Forty obstacles in a two-unit cube is dense clutter; absolute success rates are correspondingly low and should be read comparatively.

Expert demonstrations come from RRT-Connect, random shortcutting, uniform resampling to 64 waypoints, and CHOMP refinement, with dense collision validation and fallback to the raw RRT path. The dataset is 300 environments × 600 start–goal pairs × 30 trajectories, doubled by time-reversal.

B Flow-matching planner

The planner is a conditional flow-matching model over whole trajectories. A trajectory is $N = 64$ waypoints in $\mathbb{R}^3$. Training draws $\mathbf{x}_0 \sim \mathcal{N}(0, I)$ and $t \sim \mathcal{U}[0, 1]$, forms the interpolant $\mathbf{x}_t = (1-t)\mathbf{x}_0 + t\,\mathbf{x}_1$ with $\mathbf{x}_1$ an expert trajectory, and regresses the velocity target:

$$\mathcal{L} = \mathbb{E}_{\mathbf{x}_0,\mathbf{x}_1,t}\big\|v_\theta(\mathbf{x}_t, t, c) - (\mathbf{x}_1 - \mathbf{x}_0)\big\|^2, \qquad (2)$$

where c encodes the obstacles and the query. Sampling integrates $\dot{\mathbf{x}} = v_\theta(\mathbf{x}, t, c)$ from $t = 0$ to 1 by explicit Euler. The network is a dilated temporal convolutional trunk with FiLM conditioning [18, 19] fed by a PointNet-style permutation-invariant obstacle encoder [21]; 2.16M parameters throughout.

IV Method

A Decomposing the group

SE(3) has six degrees of freedom. The query supplies a frame, and that frame consumes five of them:

- **Three translations**, removed by placing the origin at the midpoint $\mathbf{o} = \frac{1}{2}(\mathbf{s} + \mathbf{g})$.
- **Two rotations**, removed by aligning the first axis with $\hat{\mathbf{x}} = (\mathbf{g} - \mathbf{s})/\|\mathbf{g} - \mathbf{s}\|$. Specifying a direction on S^2 costs exactly two degrees of freedom.
- **One rotation remains**: the roll about $\hat{\mathbf{x}}$. Having fixed which way the path runs, nothing in the problem specifies which way is "up" around that axis.

Concretely, let $d = \|\mathbf{g} - \mathbf{s}\|$, let $i^* = \arg\min_i |\hat{\mathbf{x}}_i|$, and set

$$\hat{\mathbf{y}} \propto \hat{\mathbf{x}} \times \mathbf{e}_{i^*}, \qquad \hat{\mathbf{z}} = \hat{\mathbf{x}} \times \hat{\mathbf{y}}, \qquad (3)$$

with R the rotation whose rows are $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$. A point $\mathbf{p}$ maps to $R(\mathbf{p} - \mathbf{o})$. Because the smallest component of a unit vector satisfies $|\hat{\mathbf{x}}_{i^*}| \leq 1/\sqrt{3}$, the cross product in Eq. (3) has norm $\geq \sqrt{2/3} \approx 0.816$ everywhere, so the construction is well-conditioned over the entire domain with no threshold or fallback branch, and $\det R = +1$ by construction.

Proposition 1 (SE(3) reduces to a single roll)**.** *Let X denote the reduced representation of $(\mathcal{O}, \mathbf{s}, \mathbf{g})$ and X' that of $(T\mathcal{O}, T\mathbf{s}, T\mathbf{g})$ for $T = (Q, \mathbf{t}) \in \mathrm{SE}(3)$. Then there is a single angle $\theta(T, \mathbf{s}, \mathbf{g})$ such that*

$$X' \;=\; R_{\hat{\mathbf{x}}}(\theta)\, X, \qquad (4)$$

where $R_{\hat{\mathbf{x}}}(\theta)$ is the rotation by θ about the first axis, applied to every reduced quantity — waypoints, obstacle centres and obstacle edge vectors alike. Consequently d, the first coordinate of every reduced point, and its radius in the plane normal to that axis are invariant under all of SE(3)*; the residual ambiguity is exactly one degree of freedom.*

Proof sketch. Under T the midpoint maps to $Q\mathbf{o} + \mathbf{t}$ and the direction to $Q\hat{\mathbf{x}}$, so both frames send the start–goal direction to the first basis vector. A change of basis between two right-handed orthonormal frames that agree on their first axis is a rotation about that axis, which gives Eq. (4); the invariants named are exactly the invariants of that rotation. □

It is worth being precise about what this does *not* say, because the stronger claim is tempting and false. The reduced representation is not invariant under SE(3). The gauge $\hat{\mathbf{y}}$ in Eq. (3) is built from a *fixed world axis* $\mathbf{e}_{i^*}$, so rotating the whole problem changes which axis is selected and rolls the frame. What Proposition 1 establishes is that the entire residual is that one roll — SE(3) collapses to SO(2), not to nothing. Five degrees of freedom are removed exactly, to floating-point precision, at initialisation and without any training; the sixth survives as a gauge ambiguity. This is precisely why the mechanisms of Sec. IV-C exist, and a reader who took the representation to be fully invariant would find those mechanisms unmotivated. We verify Eq. (4) numerically to 10^{-15} in both domains.

Two further consequences follow. The six numbers $(\mathbf{s}, \mathbf{g})$ collapse to the single invariant scalar d, and every pair of training examples that were "the same problem in a different pose" now coincide up to a roll, raising the effective density of the training set without generating a single new trajectory.

B Why the sixth degree of freedom is not removable

Proposition 2 (Topological obstruction)**.** *There is no continuous map $\hat{\mathbf{y}} : S^2 \to S^2$ with $\hat{\mathbf{y}}(\hat{\mathbf{x}}) \perp \hat{\mathbf{x}}$ for all $\hat{\mathbf{x}}$. More generally, for a gauge $\hat{\mathbf{y}}(\hat{\mathbf{x}}, \mathcal{O})$ that may also depend on the scene, fix any $\mathcal{O}$: then $\hat{\mathbf{x}} \mapsto \hat{\mathbf{y}}(\hat{\mathbf{x}}, \mathcal{O})$ is a tangent field on S^2, and if continuous it vanishes somewhere.*

Proof. Such a map is a continuous nowhere-vanishing unit tangent vector field on S^2, whose existence the hairy ball theorem denies [17]. The second statement is the first applied to the section at fixed $\mathcal{O}$. ☐

The second form is what the scene-derived gauges of Sec. IV-B need, and it is worth being precise about what it gives. Such a gauge is a function of $(\hat{\mathbf{x}}, \mathcal{O})$, so the bare hairy ball theorem — a statement about maps out of S^2 alone — does not apply to it. Fixing the scene repairs that, at the cost of a weaker conclusion: the degenerate set is *per scene*. Every scene admits start–goal directions at which its gauge breaks down, but which directions those are moves as the scene moves, so no restriction of the query distribution can avoid them uniformly.

Every deterministic gauge therefore carries a discontinuity; one may relocate it but never remove it. The escape route of restricting to a contractible subdomain is closed here: the minimum separation 1.5 along an axis of extent 2.0 still admits axis-aligned start–goal directions, so the domain is the whole sphere. Our i^* tie-break is exactly such a relocated discontinuity, placed on a measure-zero set.

Proposition 3 (Distributional obstruction)**.** *Let the obstacle-centre law be invariant under the central inversion $\mathbf{p} \mapsto -\mathbf{p}$, let φ_j be the angles of the centres in the plane normal to $\hat{\mathbf{x}}$, and let the weights w_j depend only on radius. Then the angular moments $c_m = \sum_j w_j e^{im\varphi_j}$ satisfy*

$$\mathbb{E}[c_m] = 0 \quad \textit{for every odd } m \textit{ and every } \hat{\mathbf{x}} \in S^2. \tag{5}$$

Proof. Inversion fixes every radius and hence every w_j, and maps $\varphi_j \mapsto \varphi_j + \pi$, so $e^{im\varphi_j} \mapsto (-1)^m e^{im\varphi_j}$ and $\mathbb{E}[c_m] = -\mathbb{E}[c_m]$ for odd m. ☐

The hypothesis is deliberately weaker than our benchmark: uniform-on-a-cube is one centrally symmetric law, and the proof covers Gaussian clutter, spherical shells, and any symmetric mixture equally. Nothing here is a theorem about our own generator.

The restriction to odd m is not a weakness of the proof but a fact about the geometry, and it is worth stating because the natural stronger claim is false. One is tempted to argue that the cube is C_4-symmetric, so only multiples of four survive. That holds when $\hat{\mathbf{x}}$ is a four-fold axis of the cube, and fails otherwise, because the plane normal to a general $\hat{\mathbf{x}}$ is not a coordinate plane and the cube's symmetry group does not act on it as C_4. Numerically, at 2×10^5 scenes (noise floor 0.002), $|\mathbb{E}[c_2]|$ is 0.001 along a coordinate axis but 0.033 for a generic direction and 0.070 along a face diagonal, while $|\mathbb{E}[c_1]|$ is at the noise floor for all of them. Eq. (5) is what the geometry actually supports.

Eq. (5) is an *unconditional* statement, and the gauge is used conditionally — a point that matters, and that closes the route more firmly than the proposition alone. A scene-derived gauge is computed in the reduced frame, i.e. from the displacements $\mathbf{p}_j - \mathbf{o}$ with $\mathbf{o} = \frac{1}{2}(\mathbf{s} + \mathbf{g})$. Conditional on the query, the law of $\mathbf{p}_j - \mathbf{o}$ is the obstacle law shifted by $-\mathbf{o}$, which is centrally symmetric about $-\mathbf{o}$ rather than about the origin. So

$$\mathbb{E}[c_1 \mid \mathbf{s}, \mathbf{g}] \neq 0 \quad \text{whenever} \quad \mathbf{o} \neq \mathbf{0}, \tag{6}$$

with the surviving moment pointing along the projection of $-\mathbf{o}$ onto the plane normal to $\hat{\mathbf{x}}$. Eq. (5) recovers $\mathbb{E}[c_1] = 0$ only after averaging over queries, since $\mathbb{E}[\mathbf{o}] = \mathbf{0}$. We confirm Eq. (6) numerically: at $\mathbf{o} = (0, 0.4, 0.3)$ the conditional first moment has magnitude 0.56 and argument 2.208, against a predicted 2.214.

That is the sharper closing of the scene-derived route. Conditioned on a query with $\mathbf{o} \neq \mathbf{0}$ the first-moment gauge is *not* degenerate — it is well conditioned, and it points from the query midpoint towards the centre of the workspace. But that direction is not an SE(3)-invariant of the problem: it is a function of where the query sits in the world. A gauge built on it would reintroduce exactly the world-frame dependence Proposition 1 removes, forfeiting five degrees of freedom to fix the sixth. The first-moment gauge is therefore either degenerate (unconditionally, at $\mathbf{o} = \mathbf{0}$) or not equivariant (conditionally, elsewhere), and neither is usable.

A gauge built from an even moment escapes Eq. (5) — inversion does not kill even m — and also escapes Eq. (6), since an $m = 2$ moment is insensitive to the sign of the displacement. It does not escape Proposition 2 in its second form: an $m = 2$ moment defines a *director* rather than a vector, and a continuous director field on S^2 is obstructed for the same Euler-characteristic reason as a vector field, since $\chi(S^2) = 2 \neq 0$. The conclusion is again per scene: for each fixed $\mathcal{O}$ there are directions at which the director degenerates. Both routes to a canonical roll are closed, for unrelated reasons.

C Handling the sixth by symmetry

Roll augmentation. During training a uniform $\theta \sim \mathcal{U}[0, 2\pi)$ is composed into R, so reduction and roll are applied as a single rotation. The prior $\mathbf{x}_0$ is drawn *after* this rotation and never rotated: an isotropic Gaussian is already rotation-invariant, so rotating it is a redundant step capable only of introducing error.

Frame averaging. At inference the field is evaluated at K rolled copies of the state, each result rotated back, and averaged:

$$(\mathcal{A}f)(\mathbf{x}) = \frac{1}{K} \sum_{k=0}^{K-1} \rho(\theta_k)^{-1} f\big(\rho(\theta_k)\mathbf{x}\big), \quad \theta_k = \frac{2\pi k}{K}. \tag{7}$$

Proposition 4 (Exact C_K equivariance and projection)**.** *For any f, $\mathcal{A}f$ is exactly C_K-equivariant. Moreover $\mathcal{A}$ is linear, idempotent, and self-adjoint with respect to $\langle f, g\rangle = \mathbb{E}_{\mathbf{x}}\left[f(\mathbf{x})^\top g(\mathbf{x})\right]$ for any C_K-invariant law on $\mathbf{x}$; hence it is the orthogonal projection onto the subspace of C_K-equivariant fields.*

Proof sketch. Rolling the input by θ_j permutes the quadrature points, and reindexing a finite sum is a bijection on them, giving equivariance and $\mathcal{A}^2 = \mathcal{A}$. Self-adjointness follows by substituting $\mathbf{x} \mapsto \rho(\theta_k)\mathbf{x}$ in the expectation, which the invariance of the law leaves unchanged, together with $\rho(\theta_k)^\top = \rho(\theta_k)^{-1}$. □

We claim no novelty for Proposition 4 or for Corollary 5 below. That group averaging is the orthogonal projection onto the equivariant subspace, and that a function decomposes orthogonally into equivariant and anti-equivariant parts, are established results in the analysis of equivariant learning [7, 20]; we restate them because the bound we are proposing is read off them. The contribution here is the measurement — the size of the anti-equivariant component of a trained planning field, and what it implies for the mechanisms built to remove it — rather than the projection theorem.

Two properties of Eq. (7) matter in practice. The guarantee requires no architectural constraint and no training assumption — it is a property of the operator, and holds for an untrained network. And because the obstacle encoding is time-independent, the K rolled scene encodings are computed once outside the ODE loop and reused at every integration step, which is why K=9 costs 1.05× the wall-clock time of K=1 (Sec. V-M).

D The non-equivariance residual

Proposition 4 turns a qualitative question — "is symmetry worth enforcing here?" — into a measurable one. Write $v_k = \rho_k^{-1} f(\rho_k \mathbf{x})$ and $\bar{v} = \mathcal{A}f(\mathbf{x})$, and define

$$r \;=\; \frac{\left(\mathbb{E}_{\mathbf{x},k}\big\|v_k - \bar{v}\big\|^2\right)^{1/2}}{\left(\mathbb{E}_{\mathbf{x}}\big\|\bar{v}\big\|^2\right)^{1/2}}. \tag{8}$$

This is the relative norm of the component of the learned field that lies outside the equivariant subspace. The norms must be the Hilbert norms of Proposition 4, i.e. root-mean-square rather than mean-of-norms; Jensen's inequality separates the two, and only the form in Eq. (8) makes Corollary 5 an identity rather than an approximation.

Corollary 5 (Symmetrisation budget)**.** *Let f^* be the target velocity field, which is C_K-equivariant because the planning problem is. Since $\mathcal{A}$ is the orthogonal projection onto the equivariant subspace and $\mathcal{A}f^* = f^*$, the error splits orthogonally:*

$$\|f - f^*\|^2 \;=\; \underbrace{\|\mathcal{A}f - f^*\|^2}_{\textit{equivariant error}} \;+\; \underbrace{\|(I-\mathcal{A})f\|^2}_{= r^2\|\mathcal{A}f\|^2}. \tag{9}$$

Hence (i) *the squared error that any equivariantisation of f can remove is at most $r^2\|\mathcal{A}f\|^2$, and $\mathcal{A}$ attains it;* (ii) *the remaining error $\|\mathcal{A}f - f^*\|$ is equivariant error, which no symmetrisation of f reduces; and* (iii) *every map sending f to an equivariant field moves it by* at least *$\|f - \mathcal{A}f\|/\|f\| = r/\sqrt{1+r^2}$, with equality only for $\mathcal{A}$.*

Part *(iii)* is the direction the projection property actually gives, and it is worth separating from *(i)*: $\mathcal{A}f$ is the *nearest* equivariant field, so symmetrising is a lower bound on how much f is disturbed, never an upper one. What is bounded above is the *error* that symmetrising can remove, and only because the target is equivariant. Conflating the two — as an earlier version of this corollary did — reverses the inequality.

The practical reading is narrower than a decision rule, and worth stating as such. Measure r; if it is small, then by Eq. (9) the model's error is overwhelmingly *equivariant* error, and symmetrising f — by frame averaging, or by any other projection onto the equivariant subspace — can remove at most an r^2 fraction of the field's squared magnitude from it. At the measured $r \approx 0.017$ that is under 3×10^{-4}, while the model is failing on more than half of its samples: the two quantities are not the same order, and no rearrangement of the symmetry can close the gap. The measurement costs one forward pass and no retraining.

What it does not license is a verdict on building an equivariant architecture, and on this point we can do better than a caveat. Symmetrising a trained field and training a constrained model are different operations: the first is a projection, bounded by Eq. (9); the second changes the hypothesis class, and with it optimisation, capacity allocation and generalisation, none of which r measures.

We test that reading directly, and it fails: see Sec. V-E, where two arms with indistinguishable residuals differ by 28 points of held-out performance.

This is also a consistency check on Proposition 1 for a *trained* model. For the reduced arm the SE(3) residual is 0.0186 at K=9 and 0.0194 at K=32, against 0.0191 and 0.0192 for the roll residual at matched K: a rigid motion acts on the reduced representation only through a roll, as claimed, and the two independently computed quantities agree to 0.0005. Table 5 quotes K=8, which is why its entry for this arm (0.0174) differs slightly from the K=9 and K=32 values here; the estimator is converged to within 0.002 over that range.

We state the limit of this argument explicitly, because it is the objection a reader should raise. Corollary 5 bounds what can be gained by symmetrising *this* model. It does not bound what an equivariant architecture could achieve, since such an architecture searches a different hypothesis class and may converge to a different — possibly better — equivariant field. What a small r does establish is that the model's residual error is overwhelmingly *within* the equivariant subspace, so *projecting this model onto that subspace* can remove very little of it. It does not follow that a constrained architecture would gain little: Sec. V-E exhibits two arms that satisfy the constraint equally well and differ by 28 points, so membership of that subspace is not what separates them.

E Implementation

Three changes make the reduction representable, and one class of error had to be designed out.

Oriented boxes. The reduction is a general SO(3) rotation, under which an axis-aligned box becomes oriented. The usual six-number representation (centre, half-extents) cannot express this. We use twelve numbers: a centre and three half-edge *vectors*, which in the world frame are $(h_x, 0, 0)$, $(0, h_y, 0)$, $(0, 0, h_z)$, so the re-encoding is lossless. The environment SDF is unchanged, since collision checking happens in the world frame after the inverse transform.

Points and free vectors transform differently. This is the error the implementation is principally designed to prevent, because it fails silently. Waypoints, endpoints, and obstacle centres are *points* and take the affine map $R(\mathbf{p} - \mathbf{o})$; the box half-edges are *free vectors* and take the rotation only, $R\mathbf{v}$. Applying the affine map to a half-edge adds $-R\mathbf{o}$ to every edge — identically — so box *position* survives while box *extent* is swamped by a common offset. The model goes blind to obstacle size while the loss curve stays healthy. The severity scales with $\|\mathbf{o}\|$, so the defect is invisible on exactly those problems whose endpoints straddle the origin.

Verification. Eighteen tests cover deliberately disjoint failure classes. Asserting $\mathbf{s}, \mathbf{g} \mapsto (\mp d/2, 0, 0)$ catches translation errors, axis ordering, and R versus $R^\top$, but is blind to handedness, since a reflection in the y–z plane fixes the x-axis pointwise; $\det R > 0$ covers that. Neither detects the point/vector confusion, so a third test asserts $\text{featurise}(\rho \cdot S) = \text{typed\_rotate}(\text{featurise}(S), \rho)$ with the reference side written as explicit loops rather than the batched operations under test, so it catches indexing and reshaping errors instead of agreeing with them. A separate suite verifies Eq. (7) against a deliberately *untrained* network — the hard case, since Proposition 4 should not depend on f.

One subtlety is worth recording for anyone measuring r. The group acts on the whole input, scene and state together. An early version of our equivariance test rolled only the state while leaving the conditioning in its original frame, and reported a large violation from a correct implementation. A diagnostic written that way manufactures the non-equivariance it claims to find.

V Experiments

A Protocol

All results use environments 250–299, which no model was trained on, evaluated as 50 environments × 10 start–goal pairs × 20 samples with 8 Euler steps. The two arms are:

- **Control**: world frame, conditioned on the raw pair $(\mathbf{s}, \mathbf{g})$.
- **Treatment**: the $(\mathbf{s}, \mathbf{g})$ reduction with uniform roll augmentation, conditioned on the invariant scalar d.

Both arms use the twelve-dimensional oriented-box representation, so it is not a confound; both train on identical trajectories for 20 epochs.

Validation split. Within training environments, validation holds out whole start–goal pairs. A per-trajectory split leaks badly here: the 30 trajectories per pair are near-duplicates (within-pair standard deviation 0.054 against an overall 0.426, because CHOMP is a local optimiser and converges to the same route from similar initialisations). Under a per-trajectory split a held-out trajectory has a training neighbour at RMS distance 0.030; under the pair split this rises to 0.162.

Two success metrics. A sample is *free* if the whole path is collision-free. We report the per-sample rate throughout, because it is the quantity the ablations compare and it is not inflated by drawing more samples. The deployment metric is different: a generative planner is run as best-of-N behind a collision check, so what a user experiences is the fraction of *queries* for which at least one of N samples is free. We report that separately in Sec. V-B, where it can be compared against a baseline that is also stochastic. Mixing the two is the main way this kind of result gets overstated: at $N = 20$ they differ by tens of points.

Statistics. We treat the number of *distinct problems* (500) as the sample size rather than the number of trajectories (10,000), since samples within a problem and problems within an environment are correlated. This gives a standard error of ≈ 2 percentage points, and ≈ 2.7 for a difference of two arms. We checked that assumption rather than asserting it: a bootstrap that resamples whole *environments* — the outermost correlated unit — gives a standard error of 2.2 points on the 15.6% straight-line rate and 2.9 on the 74.0% TrajOpt rate (10,000 resamples), so the nominal figure is if anything slightly conservative at the rates that matter here.

Loss is not comparable across arms. The treatment regresses targets in the reduced frame, where trajectories are canonicalised along $+\hat{\mathbf{x}}$, so its target variance and hence its MSE are mechanically smaller. All reported quantities are world-frame task metrics.

B Calibration: floors and ceilings

A collision-free rate on a new benchmark means nothing without a scale, so we built one. Table 1 and Fig. 1 report four classical planners and a trivial baseline on the identical 500 problems, using the same collision checker. Classical planners are given one query at a time and run single-threaded on one CPU core.

Two facts frame everything that follows.

The floor. A straight line from start to goal is collision-free on 15.6% of these problems — the clutter is dense, but not so dense that the direct path is

Planner	Free (%)	Clearance	s/query
straight line	15.6	0.055	< 0.001
TrajOpt	74.0	0.026	0.352
CHOMP	88.8	0.019	0.394
RRT-Connect	99.6	0.017	0.284
expert (RRT+CHOMP)	100.0	0.037	0.342
flow, world frame	15.4	-0.070	0.067
flow, $(\mathbf{s}, \mathbf{g})$ reduction	**45.5**	-0.010	0.067

Table 1. The same 500 held-out problems, solved five other ways. Classical timings are one CPU core and one query; the learned rows are one GPU and 20 samples per query, so the time column is not a like-for-like comparison. Classical clearance is averaged over solved problems; the learned rows are per-sample rates at 250 training environments

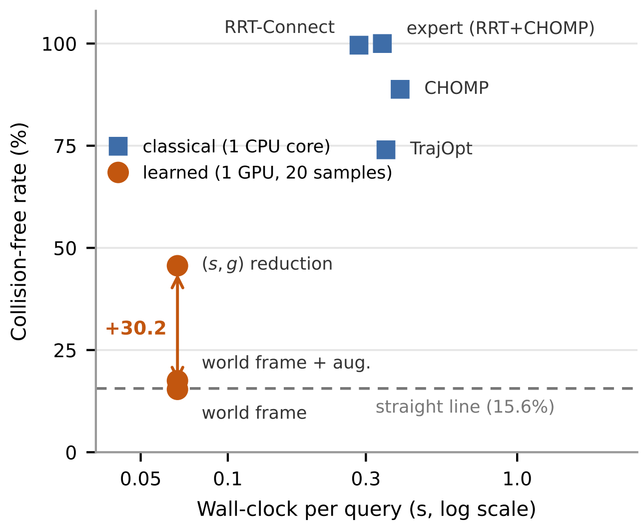

Figure 1. Where the learned planner actually sits. The world-frame arm lies on the straight-line floor; the reduction lifts it by 30.1 points; the classical planners remain far above. We show this rather than omit it: the contribution here is the representation result, not a competitive planner

usually blocked. The world-frame model reaches 15.4% per sample. The two are within 0.2 points, an order of magnitude inside the standard error. On a per-sample basis, whatever the world-frame arm has learned is worth nothing over connecting the endpoints with a line. This is the sharpest available statement of the problem the reduction solves, and it is invisible without the trivial baseline.

Augmentation is not the same lever. The comparison above shows what *ignoring* the symmetry costs. The comparison a practitioner who already knows about the symmetry would want is against the other standard way of using it: augmenting the training distribution with random rigid motions of the whole problem. We ran that arm at all four scales (Fig. 2, Table 3). It helps, it helps more as data grows — augmentation needs data to exploit — and it is not close: 12.7, 13.8, 15.3, 17.5 against the reduction's 19.2, 34.4, 40.6, 45.5. At the largest scale augmentation captures about 7% of what canonicalisation does, and leaves the world-frame arm within 2 points of the straight-line floor. Making equivariance *likely* through the data is a different and much weaker mechanism than removing the degrees of freedom outright.

Prior width σ	0	0.05	0.10	0.30
per-sample free (%)	15.6	11.5	7.4	0.4
best-of-20 (%)	15.6	**15.8**	13.4	2.2
mean path length	1.79	3.31	5.86	16.85

Table 2. The untrained stochastic baseline: Brownian bridges pinned at the endpoints, 20 samples per query, same collision check. Widening the prior buys nothing at any setting, so best-of-N over random detours is not what a learned sampler is competing against

The stochastic floor. A straight line has one sample, so it cannot be compared against a best-of-N number. The honest control there is the model's own prior with the learning removed: Brownian bridges pinned at start and goal, drawn $N = 20$ times, filtered by the same collision check. We sweep the prior width and report the *best* setting, so the baseline is steel-manned (Table 2). Perturbing the straight line does not help. The best best-of-20 result over the whole sweep is 15.8% at $\sigma = 0.05$, against 15.6% for the unperturbed line: random detours leave the workspace faster than they route around anything, and by $\sigma = 0.3$ the mean path is nine times longer than the direct one and solves 2.2%. The $\sigma = 0$ row also reproduces the straight-line rate to the digit through an independent code path, which is a check on both harnesses.

This changes the reading of the world-frame result in a way worth stating carefully. Per sample, that arm is at the straight-line floor. Per query at $N = 20$ it is not: the world-frame model trained on 20 environments solves 44.8% of queries with at least one free sample, against 15.8% for the best untrained prior at the same N. So the world-frame model has learned something real — its samples are diverse in an informative way rather than a random one — but it has not learned to make any *individual* sample better than a straight line. The reduction is what closes that second gap.

The ceiling. RRT-Connect solves 99.6% and the expert pipeline 100%, in under 0.4 s of single-core CPU time; CHOMP from a straight-line initialisation solves 88.8%. The learned sampler at 45.5% is not competitive with any of them. We report this plainly because the alternative — comparing a learned planner only against other learned planners — is how a benchmark stops measuring anything. The contribution of this paper is that the representation change is worth 30 points while the mechanisms built for the residual symmetry are worth at most a few — one indistinguishable from zero, the other unresolved. That contribution does not require the planner to win, and we do not claim it does.

C The reduction scales; the baseline does not

Table 3 and Fig. 2 give the principal result. At 250 environments the reduction reaches 45.5% against 15.4%, a gap of $\mathbf{+30.08 \pm 0.38}$ points, measured across three

Training envs	Collision-free (%) World	+aug.	Reduction	best-of-20 Reduction
20	12.9	12.7	19.2	57.8
60	13.6	13.8	34.4	79.6
150	14.7	15.3	40.6	85.6
250	15.4	17.5	**45.5**	**90.4**

Table 3. Held-out performance against training-set size. "+aug." is the world-frame arm trained with random SE(3) motions of the whole problem. The world-frame arm gains 2.5 points across a 12.5× increase in environments and augmentation adds at most 2.1 more, while the reduction gains 26.2 and has not plateaued. Every world-frame and reduction cell is a mean over three seeds; the seed-matched differences are $+6.37 \pm 0.19$, $+20.87 \pm 1.56$, $+25.85 \pm 0.37$ and $+30.08 \pm 0.38$. The "+aug." column is single seed, so it is not compared against a three-seed mean anywhere. The last column is the deployment metric, best-of-20 behind a collision check

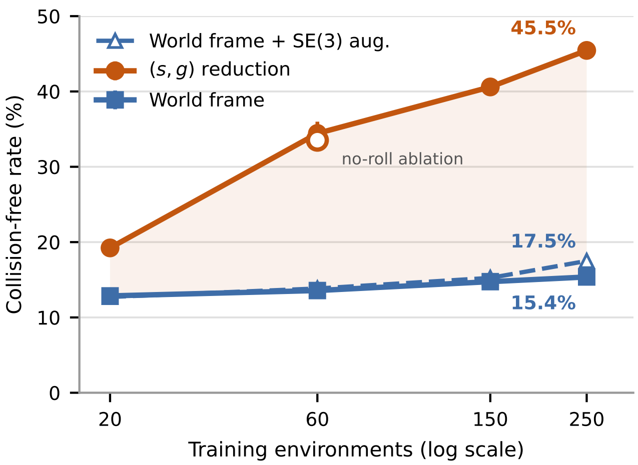

Figure 2. The gap widens with data. If canonicalisation were compensating for a shortage of training data one would expect it to close; instead the world-frame baseline is nearly flat while the reduced model continues to improve. The hollow marker is the no-roll ablation, drawn on the treatment line because it is the same arm with one component removed

seeds per arm.

The uncertainty is quoted from the seed spread in the same cell, which is what makes it a test rather than an extrapolation. Both arms were run at seeds $0, 1, 2$ at every scale, so the differences are seed-matched throughout: $+6.37 \pm 0.19$, $+20.87 \pm 1.56$, $+25.85 \pm 0.37$ and $+30.08 \pm 0.38$ at 20, 60, 150 and 250 environments. The 250-environment gap is roughly 79 across-seed standard errors. We note that the seed spread is not constant across the grid — the reduction's spread at 60 environments (1.56) is an order of magnitude larger than at 250 (0.11) — which is precisely why we do not transport a standard error from one cell to another, and why an earlier version of this paper, which quoted the 60-environment spread against the 250-environment difference, overstated the test it was performing.

The trend is more informative than the endpoint. The

Mechanism	20 ep.	converged	Assessment
$(\mathbf{s}, \mathbf{g})$ reduction (5 DOF)	+30.1	+36.5	± 0.4 / ± 0.3, n=3
of which: 3 translations	+12.3	—	recentring alone
of which: 2 rotations	+18.0	—	aligning the axis
SE(3) augmentation (world frame)	+2.1	—	real, but 7% of the ab
Roll augmentation	+0.8	—	± 0.3 paired, n=3; n.s.
Frame averaging (K:1→3)	+0.1	+0.68	± 0.13 converged, n=3

Table 4. Decomposition by mechanism on the same held-out set, in percentage points against the arm without that mechanism — so the rows are not additive, and the two indented rows decompose the row above them. Canonicalisation accounts for the effect at either budget, and both of its parts matter: recentring and axis alignment contribute 41% and 59%. The two columns are the same mechanisms measured at 20 epochs and at convergence; reading only the first would put frame averaging at zero, which Sec. V-D shows is an artefact of the budget rather than a property of the mechanism. Dashes are comparisons we did not run to convergence. The SE(3)-augmentation and translation-only arms are single seed and are therefore differenced against the SAME seed, never against a three-seed mean

baseline improves by 2.6 points across a 12.5× increase in training environments: it is close to unable to exploit additional layouts, because each new layout arrives at poses it must learn about separately. The reduction improves by 26.2 points over the same range and has not levelled off. This is the opposite of what a small-data-crutch account predicts, under which the advantage should shrink as data accumulates. Minimum clearance corroborates through an independent continuous measure: the treatment moves from -0.058 toward -0.0095, approaching feasibility, while the baseline is static.

Endpoint error falls by a factor of three to five under the reduction ($0.0101 \to 0.0041$ at 20 environments, $0.0059 \to 0.0020$ at 250), a direct consequence of canonicalisation: with endpoints always at $(\mp d/2, 0, 0)$ the model no longer spends capacity on *where* they are.

D Frame averaging's value scales with model quality; the roll ablation is unresolved

Table 4 and Fig. 3a decompose the result. Removing roll augmentation (the reduction with a fixed gauge) costs 34.4% against 33.5% at 60 environments. Both arms were run at the same three seeds, so the comparison should be paired, and paired it is +1.2, +0.2 and +1.1 points: a mean of $+0.8 \pm 0.3$, consistently positive but $t = 2.6$ on two degrees of freedom, which does not clear significance at n=3. Unpaired, the across-arm spread swamps it entirely (0.4 SE). Frame averaging on the best model gives 45.63% at K=1 and 45.70% at K=3 — 0.05 across-seed standard errors.

The reduction itself decomposes. A translation-only variant, which recentres on the query midpoint without rotating, reaches 27.7% at 250 environments, between the world frame's 15.4% and the full reduction's 45.6% at the same seed. So the three translational degrees of freedom are worth +12.3 points and the two rotational ones a further +18.0: 41% and 59% of the total. Neither

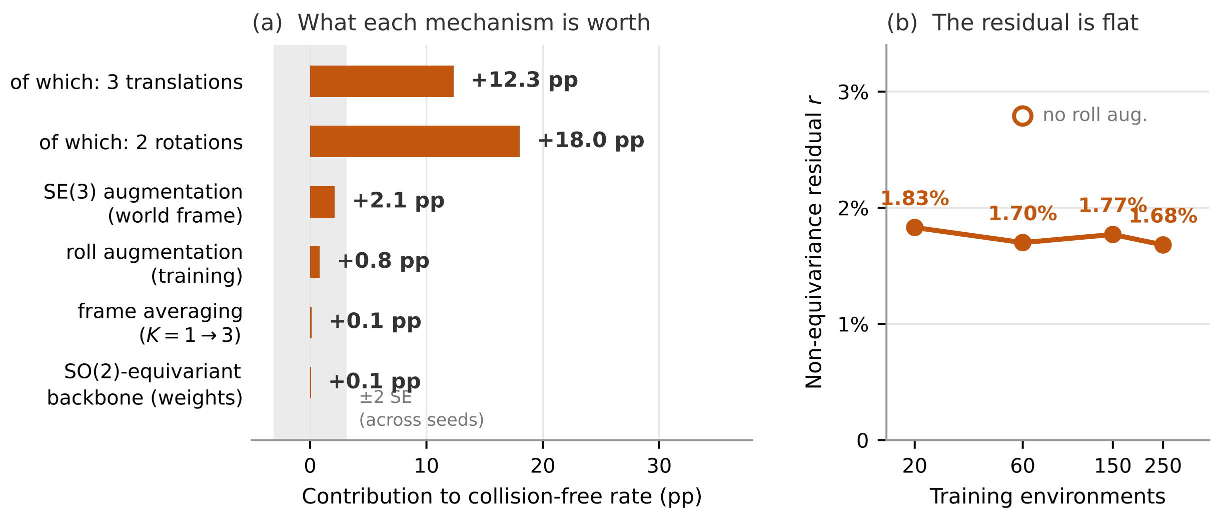

Figure 3. (a) Canonicalisation is seven times the next largest mechanism. The band is ±2 across-seed standard errors, measured on the treatment arm at 60 environments; frame averaging sits inside it, and the roll-augmentation bar rests on a single-seed ablation. (b) The non-equivariance residual of Eq. (8) is flat across a 12.5× range of training data — it is not something the model acquires from seeing more layouts. The hollow marker shows that removing roll augmentation raises the residual by 64% but leaves it under 3%

half carries the effect on its own, which is worth knowing — recentring is the trivially easy part of the reduction, and it accounts for two fifths of the benefit.

Rather than infer that from flat curves, we measure r. On the best model $r = 0.0168$ at $K{=}3$: under 2% of the field is non-equivariant. By Eq. (9) the squared error any equivariantisation can remove is at most r^2 of the field's squared magnitude, i.e. under 3×10^{-4}, while the model is failing on more than half of its samples. At *this* budget the remaining error is overwhelmingly equivariant error, which no symmetrisation reaches.

But the budget is doing the work, not the mechanism. Scoring every snapshot of the convergence runs at both $K{=}1$ and $K{=}3$ costs nothing extra and turns that single number into a curve. Across three seeds at 60 environments the gain rises from +0.19 early to $+0.68 \pm 0.13$ over the final eight snapshots, positive in every seed, while r sits at 0.0180–0.0183 throughout. At 250 environments it is steeper still: +0.22 at epoch 10, peaking at +1.57 near epoch 180, settling around +0.9, with r moving only from 0.0163 to 0.0173. The mechanism the 20-epoch grid cannot distinguish from zero is worth roughly seven times that on a converged model.

That is what Corollary 5 predicts once it is read as a statement about *proportions*. The anti-equivariant fraction of the field does not change with training, but the total error falls, so the share of the remaining error lying outside the equivariant subspace — the only part symmetrisation can reach — grows. The honest claim is therefore not that frame averaging is inert, but that its value scales inversely with how much equivariant error is left. At our main grid's budget there is far too much of it for the r^2 ceiling to matter; on a better model it begins to. The ordering of Table 4 is unchanged either way — +0.68 against the reduction's +36.5 — but "inert" is the wrong word.

The statistics here are stronger than elsewhere in the paper, and it is worth saying why: $K{=}1$ and $K{=}3$ are evaluated on the same weights, the same problems and the same prior draws, so the difference is paired within a checkpoint and carries no seed variance at all. The across-seed standard error of 1.56 points, which governs comparisons between separately trained arms, is simply the wrong noise model for it.

It is quality that drives this, not r. The two are easy to conflate, so we separated them. Along the convergence curve r is nearly constant while quality varies by 17 points and the benefit grows sevenfold. To vary r instead, we restricted training to a cone of start–goal directions about the first axis, starving the reduced-frame data of the implicit roll diversity that makes the field near-equivariant in the first place. The manipulation works: at cone half-angles of 90, 45, 30 and 15 degrees, with training-set size and gradient-step count held fixed, r rises monotonically through 0.0182, 0.0192, 0.0202, 0.0238. The frame-averaging benefit over the same sweep is +0.02, −0.05, +0.44, −0.25: scatter about zero, with the largest r giving the most negative gain.

We report that sweep as an attempted decoupling that *failed*, because it is confounded and saying so is more useful than the conclusion it appears to support. Narrowing the cone also costs generalisation — $K{=}1$ falls from 31.4 to 20.6 — so r and quality move in opposite directions by construction, and a flat result is consistent both with r being irrelevant and with r mattering while quality cancels it. Re-scoring every arm on a common 15-degree cone of held-out queries, a subset of all four training cones so that no arm extrapolates, lifts them all by about 15 points and confirms that much of the collapse was distribution shift — but it does not equalise them: the spread stays at 12 points, because an arm trained on a narrow cone generalises worse even *within* that cone despite seeing the same number of trajectories. And among the three arms whose quality does match (45.0, 46.0, 46.9), r barely varies (0.0195, 0.0196, 0.0187). The only setting that moves r appreciably is the one that destroys quality, so on this benchmark the cone dial cannot separate them. Read *with* the convergence curve, where r is fixed and quality alone varies, the variable that tracks the benefit is quality.

Two observations sharpen this. First, removing roll augmentation raises the residual by 64% ($0.0170 \to 0.0279$ at 60 environments) — a large relative change, but between two values that are both under 3%, so augmentation is not what makes the field near-equivariant in the first place. The explanation is that the gauge is a deterministic function of $\hat{\mathbf{x}}$, and 600 start–goal pairs per environment supply 600 distinct directions; the reduced-frame data already spans a wide spread of rolls, and explicit augmentation was solving a problem the data had already solved. Second, and contrary to what we expected, the residual does *not* move with scale: 0.0183, 0.0170, 0.0177, 0.0168 across a 12.5× range of training data (Fig. 3b). It is small and flat. An earlier two-point reading under a mean-of-norms definition of r suggested a downward trend; measured at four scales under Eq. (8) it is a constant, and we report it as one.

That the residual is flat rather than falling is itself informative. It means the near-equivariance of the learned field is not something the model acquires gradually from seeing more layouts — it is present at 20 environments and unchanged at 250. Combined with the no-roll ablation, the picture is that the reduced-frame data supplies enough implicit roll diversity from the outset, and that nothing about the scale of the training set changes how equivariant the learned field is. What rests on the *size* of r and Corollary 5 is the narrower conclusion that *symmetrising this model* cannot help, and that conclusion does not depend on a trend. It says nothing about what a constrained architecture would achieve, for the reason Sec. V-E makes concrete.

E What the residual does not predict

To test the tempting reading directly we measured r under the *full* SE(3) action rather than the roll alone, which makes it defined for the world-frame arm as well: K random rigid motions are applied to the whole problem, the field is evaluated in each, and each result is mapped back by the inverse rotation. Table 5 is the result.

Arm (250 environments)	r under SE(3)	Free (%)
World frame	0.0881	15.4
World frame + SE(3) augmentation	0.0181	17.5
$(\mathbf{s}, \mathbf{g})$ reduction	0.0174	**45.6**

Table 5. The residual does not predict what a mechanism is worth. Augmentation and canonicalisation leave the learned field equally equivariant, and differ by 28 points of held-out performance. Both columns are read from the SAME seed-0 checkpoints, since r is a property of a particular trained field; the three-seed mean for the reduction row is 45.5

Augmentation drives the residual down by a factor of 4.9: it demonstrably teaches the network the symmetry, to the same degree the reduction enforces it by construction. It buys 2.1 points. The reduction, at an indistinguishable residual, buys 30.1. *Two models with the same r can differ by* 28 *points*, so r does not forecast the benefit of a symmetry mechanism, and a decision rule of the form "build the constrained architecture only if r is large" has no support here. What r bounds is what symmetrising a given field can remove, which is Eq. (9) and is a theorem; what it does not measure is whether symmetry is the binding constraint at all.

One number in Table 5 deserves saying out loud. The world-frame arm ignores the symmetry entirely and performs at the straight-line floor — and it is still 91% equivariant. A model that has learned nothing useful about planning has nonetheless landed close to the equivariant subspace, because most of what a velocity field does on this problem is roughly pose-independent regardless of how it was trained. That is the sharpest form of the point: r being small is compatible with having solved the problem, with having learned the symmetry and nothing else, and with having learned nothing at all.

F The third mechanism: a constrained architecture

Sec. IV-C distinguishes three places a symmetry can be imposed: in the data, in the operator, or in the weights. The first two are measured above. Earlier versions of this work stated the third as untested, and were careful not to let Corollary 5 stand in for it: the bound governs post-hoc projection of a *given* field, whereas a constrained architecture searches a different hypothesis class. We now report it.

We built a backbone that is exactly SO(2)-equivariant about the start–goal axis by construction. Features are separated by irreducible representation — the component along the axis is invariant ($m = 0$), the pair in the normal plane rotates ($m = 1$) — and only operations respecting that separation are permitted: complex-linear channel mixing, obstacle pooling with attention weights computed from invariants alone, normalisation of the rotating stream by an invariant, and gating of the rotating stream by scalars. The property is verified on an *untrained* network, which is the hard case and the only one that shows equivariance comes from the structure of the weights rather than from training: the equivariance error is 1.2×10^{-6}, and it survives multiplying every parameter by three. The same test reports 5.03 for a rotation off the axis, so a network that ignored its vector inputs entirely — which would be trivially equivariant — fails it.

Capacity was not matched downward. At the

Epoch	Equivariant	Unconstrained	Δ
10	35.4	22.1	+13.3
40	44.6	37.9	+6.7
100	48.7	45.8	+2.9
300	51.3	51.1	+0.05

Table 6. An SO(2)-equivariant backbone against the unconstrained one, 60 environments, identical data and schedule. The constraint buys training efficiency and not a higher ceiling. The final row is seed-matched over two runs; the per-seed pairs are 50.9 vs 50.8 and 51.6 vs 51.6.

60 environments	Global only	+ local	Δ
World frame	15.0	54.5	+39.5
$(\mathbf{s}, \mathbf{g})$ reduction	50.8	79.2	+28.4
+ equivariant backbone	50.9	**82.2**	+31.3
reduction contributes	+35.8	+24.7	

Table 7. Local geometry against the query frame, at convergence. All cells are seed 0 so the differences are seed-matched; the three-seed means for the two left-hand cells in the first two rows are 14.60 ± 0.20 and 51.10 ± 0.25. The right-hand column is currently a single seed.

shared hyper-parameters the constrained model carries 2,582,233 parameters against the unconstrained backbone's 2,161,283, or 19.5% more. This is deliberate: a constrained model that is also smaller confounds architecture with capacity, and the uncharitable reading is the one a reader reaches for. The comparison below is therefore against a *larger* constrained model, which strengthens rather than weakens its conclusion.

Table 6 is unambiguous in both directions. The constrained model reaches at epoch 10 what the unconstrained one needs epoch 40 to reach, and at epoch 40 what it needs epoch 90 to reach — roughly a fourfold saving in optimisation. By convergence the advantage is +0.05 points seed-matched, which is zero. The same pattern holds at 250 environments, where the constrained arm reaches 54.8 by epoch 80 against the unconstrained arm's 51.6.

This completes the taxonomy, and the three mechanisms agree. Roll augmentation is worth +0.8, frame averaging +0.68, and an equivariant architecture +0.05 at convergence. Three mechanisms that impose the same symmetry by three unrelated means all recover almost nothing, which is a considerably stronger statement than any one of them alone. It also sharpens the correct reading of Sec. V-E: r did not predict this outcome and could not have, but the outcome is what the residual's smallness would have suggested — the inference is unsound even where the conclusion happens to hold.

G Local geometry, and what the reduction is worth beside it

An obstacle encoder that max-pools 40 obstacles into one scene vector, applied identically to all N waypoints, gives a waypoint no channel through which to ask what is near *it* — while collision avoidance is exactly that question. Sec. V-I shows the reduction addresses this indirectly, by making the scene code query-dependent. It can also be addressed directly: we append to each waypoint the signed distance to the scene and its gradient there, which for the constrained backbone splits cleanly by irreducible representation (d and the axial component of ∇d are invariant, the normal component rotates). This is a featurisation, not an oracle. The signed distance is a deterministic function of the obstacle primitives the network already receives; no privileged information enters. What it does assume is that obstacles are available as explicit primitives at inference, which holds in this benchmark and fails for a planner operating on perceptual input.

Three things follow from Table 7, and we state the uncomfortable one first.

Local geometry is the larger single lever. Against the world frame it is worth +39.5 points where the reduction is worth +35.8. The representation change this paper is about is not the largest effect available on its own benchmark. We report this because it bears directly on how the contribution should be read: the claim that survives is that a query-derived frame is worth a great deal at zero cost, not that symmetry is the dominant consideration in designing a generative planner.

The reduction is not merely compensating for a weak encoder. That is the obvious objection to the paragraph above, and it is testable: if the frame were only a workaround for a scene code that cannot vary with the query, then supplying per-waypoint geometry should absorb its benefit. It does not. With local geometry present the reduction is still worth +24.7 points. We ran this cell specifically to find out whether it would overturn our main result, and report that it did not.

The two are sub-additive, and the constraint finally pays. Treating the effects as independent predicts $15.0 + 35.8 + 39.5 = 90.3$; the measured value is 79.2. Both address the same underlying difficulty — relating query geometry to scene geometry — and neither subsumes the other. Against this, the equivariant backbone is worth +0.1 points without local geometry and +3.0 with it. That interaction is what the architecture predicts: the rotating stream has nothing to act on until directional information reaches individual waypoints, and once it does, enforcing the symmetry on that stream is no longer free of consequence.

H How much of the level is training budget

Every arm above is trained for 20 epochs, and that budget is not enough. Holding the data fixed at 60 environments and training the reduced arm for 60 epochs instead raises the held-out collision-free rate from 34.4% — the mean over three seeds — to **42.4%**, with minimum clearance improving from −0.031 to −0.013 and endpoint error from 0.0024 to 0.0022. Three times the optimisation, at identical data and identical capacity, is worth +8.0 points, nearly twice the largest of the mechanisms built for the sixth degree of freedom. Against the single seed the 20-epoch cell was originally reported from, the same comparison would read +11.1; we quote the seed mean throughout.

Two consequences follow, and the second is a correction.

First, this is the direct evidence that the absolute level is a budget artefact rather than a ceiling of the method, and it is the evidence we rely on rather than the train–validation comparison discussed in Sec. VI. The model at 20 epochs is substantially underfit, so the 45.5% headline understates what this architecture reaches at convergence.

Second — and this is the part we did not anticipate — the same tripling is worth almost nothing to the world-frame arm. Running it at 60 epochs on the same 60 environments moves it from 13.6% to 14.4%:

60 environments	20 ep.	60 ep.	gain
per-sample collision-free (%)			
World frame	13.6	14.4	+0.9
$(\mathbf{s}, \mathbf{g})$ reduction	34.4	**42.4**	+8.0
best-of-20 (%)			
World frame	42.3	39.2	−3.1
$(\mathbf{s}, \mathbf{g})$ reduction	79.6	**87.2**	+7.6

Optimisation buys +8.0 points with the reduction and +0.9 without it, an interaction of +7.1 points at roughly 4.5 across-seed standard errors. The two mechanisms are not substitutes: they relieve different constraints. The representation bottleneck — the network being shown a pose it must relearn at every value — is removed by canonicalisation and *cannot* be removed by compute. The capacity bottleneck is removed by compute and cannot be removed by canonicalisation. This also settles the reading of Fig. 2: the world-frame arm's flatness is not an artefact of an insufficient budget, because giving it three times the budget leaves it flat.

The deployment metric at this budget is high: best-of-20 behind a collision check solves **87.2**% of held-out queries, against 15.8% for the best untrained prior at the same N (Table 2). A reader who cares about whether the sampler is useful rather than about the representation question should read that number, with the caveat from Sec. V-B that classical planners still solve more of them and faster.

On that metric the interaction is not merely larger but changes sign for the control. The reduction rises from 79.6% to 87.2%, while the world-frame arm *falls*, from a three-seed mean of 42.3% (spread 41.4–43.2) to 39.2%: an interaction of +10.7 points against the +7.1 measured per sample. Tripling the optimisation of the world-frame arm made it worse on the metric that describes how a generative planner is actually deployed — more training bought a per-sample gain of +0.9 while costing sample diversity, which is the resource best-of-N actually spends. Taken alone this cell would be weak evidence — it is single seed against a three-seed mean, the configuration that manufactured the roll-augmentation artefact of Sec. VI — and we do not rest it on this domain. It is corroborated independently in Sec. V-K, where the same quantity declines monotonically across an 8× budget range in a different state space, with no seed-matched comparison involved. Two domains, two budget ranges, one direction. The +7.1 per-sample interaction is separate and does not depend on either.

Trained to a plateau, the world-frame arm does not clear the straight line. The 3× comparison above answers the convergence-rate objection with a trend rather than a limit, so we ran both arms at 60 environments to 300 epochs — a 15× budget — snapshotting every 10 epochs and scoring every snapshot on the same held-out problems.

Epochs	20	60	150	300	last 10
World frame	13.2	14.4	14.2	14.60 ± 0.20	+0.1
$(\mathbf{s}, \mathbf{g})$ reduction	31.4	42.3	48.3	$\mathbf{51.10 \pm 0.25}$	+0.2
Gap	18.2	27.9	34.1	$+\mathbf{36.5 \pm 0.32}$	

Every cell is a mean over three seeds. Both arms are flat by the end — the control gains 0.1 points over the final ten epochs and the reduction 0.2 — so this is a converged comparison rather than a comparison of convergence rates. Three things follow.

The world-frame arm does not reach the straight line. It converges to 14.60 ± 0.20% against the 15.6% of Sec. V-B, measured on the identical problems — roughly five across-seed standard errors *below* it. At 20 epochs the two were indistinguishable (15.37 ± 0.36); trained to a plateau with fifteen times the budget the learned arm is worse than joining the endpoints with a line, and the extra optimisation is what makes the deficit resolvable rather than what causes it. This is the strongest form of the paper's central negative claim, and at 20 epochs it was not available: a reader could always answer "it is undertrained".

The gap *widens* monotonically with budget, from 18.2 to 36.5 points. Undertraining was not manufacturing it; if it had been, the flat arm would be the one with headroom.

And the reduction at 60 environments and 300 epochs (51.1%) exceeds the reduction at 250 environments and 20 epochs (45.5%), so epochs and environments are substitutable over this range — which bears directly on the dataset-generation argument of Sec. VI.

At full data the control gains nothing at all. Repeating the study at 250 environments sharpens it. Over 300 epochs the world-frame arm moves from 14.6% to 14.4% — it ends *below* where it began and never leaves the band 14.2–15.3 — while the reduction climbs from 38.8% to 55.8%:

250 environments	10	60	150	300	last 10
World frame	14.6	14.5	14.9	14.4	−0.2
$(\mathbf{s}, \mathbf{g})$ reduction	38.8	50.5	53.1	**55.8**	+0.6

Linear probe on the encoder output	World	Reduction
scene code, within-env. std	0.000000	0.182
as a fraction of total std	0.00	0.95
R^2, scene code → clearance	+0.015	**+0.778**
R^2, full conditioning → clearance	+0.07	**+0.878**
R^2, separation d alone	—	0.000

Table 8. What the scene code knows about the query. Ridge probes from the 128-d obstacle embedding, and from the full conditioning vector, to the minimum clearance along the straight line from $\mathbf{s}$ to $\mathbf{g}$ — the single thing a planner most needs to know about a query. Fitted and scored on *disjoint* environments, so a probe cannot memorise a scene; the conditioning row is the best score over $\lambda \in [1, 10^4]$

Obstacles	line	World	Reduction	Gap	Red. best-of-20
8	61.3	61.4	90.8	+29.4	100.0
16	41.3	44.5	79.8	+35.3	100.0
24	24.7	24.8	66.0	**+41.2**	98.0
32	18.0	18.7	53.5	+34.8	92.7
40 (trained)	11.3	13.5	46.4	+32.9	89.3

Table 9. Both arms evaluated across a 5× range of clutter, on freshly generated layouts neither model has seen. Scoring needs no expert trajectories — only obstacles, starts, goals and the collision checker — so a density axis costs minutes rather than the days a training sweep would. Both arms are trained at 40 obstacles, so every other column is out of distribution: this measures generalisation across density, not the effect of training at another one

The gap reaches **41.4** points. At the full training set, trained to convergence, a world-frame flow-matching planner is at 14.4% against a 15.6% straight line — and thirty times the optimisation buys it nothing. This is the strongest form of the claim in the paper, and at 20 epochs it was unavailable.

We are precise about what has and has not plateaued. The control is flat, and that is the claim. The reduction is still gaining 0.6 points per ten epochs at epoch 300, so 55.8% is a lower bound on what this architecture reaches and not a measured ceiling.

I Where the benefit comes from

The 30-point gap invites a mechanism, and the architecture supplies one. Recall that the obstacle encoder max-pools all forty obstacles into a single 128-d vector, which FiLM then applies *identically* to all 64 waypoints.

In the world frame the encoder's inputs do not depend on the query at all, so its output is constant across every query in an environment — measured within-env. standard deviation 0.000000, which is a fact about the architecture rather than about training. In the reduced frame the obstacles are expressed in the query's own frame, so the identical encoder produces a code that varies almost entirely within an environment (95% of its total variance).

Two things follow, and only the second is obvious in advance. Table 8 shows that the max-pool does *not* destroy the query-relevant geometry: in the reduced frame a linear probe recovers how blocked the direct path is at $R^2 = 0.78$, on held-out environments, from that single global vector. And the world-frame arm's *full conditioning* — which does contain $(\mathbf{s}, \mathbf{g})$ — cannot express the same quantity linearly across environments, topping out at 0.07 over four orders of magnitude of regularisation. The treatment's score is not the separation d in disguise: d alone scores 0.000.

The reduction therefore moves the query–scene interaction out of the network's interior and into the input representation. In the world frame that interaction must be reconstructed inside the trunk from a per-environment constant and six query numbers, delivered through a global modulation; in the reduced frame it is already present in the features. We offer this as the mechanism behind Sec. V-B's finding that the world-frame arm never exceeds the straight line, and we are explicit that it is a claim about representation content, measured with linear probes on two trained checkpoints, not a causal intervention.

It also bounds what the encoder can do at all. A single global code can say *that* a path is blocked; it cannot say *which waypoint* should move and in which direction, because every waypoint receives the same modulation. That is a candidate explanation for the level at which the reduced arm plateaus, and it is separate from the gap: both arms share the encoder. We implement an oracle diagnostic for it — appending the true SDF and its gradient at each waypoint, which measures the headroom a better encoder could reach without designing one — and report no result from it here, since running it is future work rather than a finding.

J The world-frame arm is the straight line at every density

The finding of Sec. V-B — that the world-frame arm scores what a straight line scores — is stated at one density, where it can be read as coincidence. It is not. Across a 5× range of clutter the world-frame arm tracks the straight-line floor to within a few points at every setting, and to within a tenth of a point at three of five (Table 9). The reduction clears the floor by 29 to 41 points everywhere.

The gap is not monotone in difficulty. It peaks near 24 obstacles at +41.2 and falls off in both directions, which is what a quantity bounded below by the floor and above by the ceiling should do; our training density of 40 sits on the dense side of that peak.

This also answers a question about the benchmark's design that we would otherwise have to argue rather than measure. Thinning the clutter raises both arms substantially — the reduction reaches 90.8% per sample at 8 obstacles — but the *deployment* metric saturates: best-of-20 hits 100% below about 24 obstacles, at which point the two arms are no longer distinguishable as planners even though they still differ by 29 points per sample. Forty obstacles is not an arbitrary choice of difficulty; it is roughly the density at which best-of-N still discrimi-

SE(3) rigid body, 250 envs	20 ep.	∼156 ep.	gain
per-sample collision-free (%); floor 4.8			
World frame	2.8	3.1	+0.3
$(\mathbf{s}, \mathbf{g})$ reduction	6.2	**10.0**	+3.8
best-of-20 (%)			
World frame	16.8	15.0	−1.8
$(\mathbf{s}, \mathbf{g})$ reduction	24.8	**40.6**	+15.8

Table 10. The second state space, at two budgets an 8× factor apart. The trivial baseline for this domain is geodesic interpolation — straight in position, slerp in rotation — the SE(3) analogue of joining start to goal with a line. The world-frame arm gains 0.3 points across the whole budget range and never clears its floor at any budget; the reduction clears it and is still climbing. On the deployment metric the world-frame arm moves *backwards*, as it does in the point-mass domain. Single seed per cell

nates.

K Replication in a second state space

The point-mass domain has a convenient property that could be doing the work: the state is a point, so the points-versus-vectors distinction of Sec. IV-E lives only in the obstacle features, never inside the trajectory. To test whether the result survives without that convenience we built a second domain — an L-shaped rigid body in the same clutter, where a state is a full pose, position and orientation, rather than a point. The trajectory now carries orientation, so the typing distinction moves inside the state. Collision checking evaluates the obstacle SDF at the body's transformed sphere centres; steering interpolates linearly in position and geodesically in rotation; and the trivial baseline is the corresponding geodesic interpolation, the SE(3) analogue of joining start to goal with a line.

Three things replicate (Table 10). The world-frame arm fails to beat the trivial baseline in *both* domains — 1.8 points below the geodesic floor here, 1.2 below the straight-line floor there — so the finding of Sec. V-B is not an accident of one workspace family. The reduction clears its floor in both. And at matched budgets the ratio between the arms is 2.9× in each: an intermediate rigid-body checkpoint reaches 3.0 and 8.6, against 14.4 and 42.4 for the point mass.

This domain also supplies a near-converged test of Sec. V-H, and it agrees. The SE(3) cells above span an 8× budget range rather than the 3× available in the point-mass domain, and the best-validation checkpoints sit at epochs 155 and 159, so neither arm had stopped improving on validation loss when the runs ended. Over that range the world-frame arm gains 0.3 points and the reduction 3.8: the interaction survives at a budget where the "both arms are simply undertrained" reading is much harder to sustain. We regard this as the strongest form of the two-bottleneck claim currently available, precisely because it does not rest on the 3× comparison a reader is most likely to challenge.

The best-of-20 column carries the sharper version. In the point-mass domain we observed the world-frame arm losing 3.1 points of deployment performance under 3× the budget, and flagged it as a single cell against a three-seed mean. Here it declines monotonically across the budget range, 16.8 → 15.0, with no seed-matching subtlety, while the reduction rises 24.8 → 40.6. Two state spaces agree that additional optimisation makes the world-frame arm *worse* at best-of-N — it buys a marginal per-sample gain by spending the sample diversity that best-of-N actually consumes — and that is a stronger statement than either domain supports alone.

We are deliberately careful about how much this carries. The absolute numbers are small (3.0% and 8.6% against a 4.8% floor), a rigid body in this clutter is a substantially harder problem than a point, and every SE(3) cell is single seed. The useful statement is the *pattern*, not the level: the reduction is not exploiting something peculiar to a point-shaped state, and the implementation survives having the failure mode of Sec. IV-E moved from the conditioning into the trajectory itself, which is the part most likely to break silently.

Objective	NFE	World	Reduction	Gap
20 epochs, three seeds				
flow matching	8	13.57	34.43	$+20.87 \pm 1.56$
DDPM	8	8.88	23.20	$+14.32 \pm 0.81$
DDPM	100	12.27	28.97	$+16.71 \pm 0.42$
300 *epochs*				
flow matching	8	14.60	51.10	$\mathbf{+36.50 \pm 0.32}$
DDPM	100	13.51	50.52	**+37.01**

Table 11. The generative-model control. Same backbone, same conditioning, same data, same frame options, same 20-epoch budget at 60 environments; only the objective differs. Three seeds per cell, errors across seeds. Network evaluations are matched within a row, so each column is a like-for-like comparison

L The effect is not specific to flow matching

Every learned number to this point comes from conditional flow matching, which leaves open whether the reduction is exploiting something about straight-line interpolants rather than something about the problem. We therefore ran the same two arms under a Diffuser/MPD-style DDPM objective [12, 3]: ϵ-prediction on a cosine schedule, sharing the backbone, the conditioning, the oriented-box features and the frame options unchanged, so the objective is the only difference.

The gap replicates, and the budget turns out to matter more than the objective (Table 11). At 20 epochs the DDPM gap is +16.7 against flow matching's +20.9, each at its own best integrator budget, and an earlier version of this paper reported that difference as a real if modest objective-dependence. Trained to convergence the two coincide: +37.0 for DDPM against +36.5 for flow matching, with absolute performance also within a point (50.5 against 51.1). The apparent dependence on the interpolant was an artefact of a budget at which neither objective had converged.

We regard this as the strongest form of the generality claim. The reduction is a statement about the problem, not about the generative model, and at a budget where both objectives are actually trained it pays the same.

One methodological point is worth recording, because scoring this arm carelessly would have produced the opposite reading. The 8-step operating point used throughout this paper was chosen because straight-line interpolants tolerate it (Sec. V-M). DDPM does not: it gains roughly four points on *both* arms between 8 and 100 evaluations. Reporting the DDPM arm only at 8 steps would have understated it and invited the fair objection that the baseline was handicapped by a budget tuned for the method it is meant to test. We report both, and read the comparison at each objective's own best setting.

The converged DDPM cells are single seed, against three for the 20-epoch rows; the +37.0 should be read as a point estimate whose agreement with the flow arm's seed-matched $+36.50 \pm 0.32$ is the informative part.

M Cost

Frame averaging at $K{=}9$ costs 1.05× the wall-clock time of $K{=}1$ (0.070 vs 0.067 s per query). At 64 waypoints and batch size 1 the sampler is bound by kernel launch overhead rather than arithmetic, so a ninefold increase in width is nearly free. This matters for the interpretation: the finding that frame averaging does not help is not a budget trade-off. It was affordable, and we adopted it anyway for the measurement.

Separately, reducing the integrator from 20 to 8 Euler steps costs 0.4–0.8 points and runs 1.4× faster, so 8 steps is used throughout.

VI Discussion and Limitations

What the technique contributes. Of the six degrees of freedom of SE(3), five can be removed exactly by a frame read off the query, at no computational cost and with no architectural constraint. Doing so roughly triples the collision-free rate on this benchmark and, more importantly, converts a model that cannot use additional training environments into one that can. The sixth cannot be removed and, here, does not need to be — a contingent empirical fact rather than a theorem, and one better supported for one mechanism than the other. The K sweep and the residual both bear on *frame averaging*: the residual bounds it because frame averaging is precisely the projection, so the two are less independent measurements than two views of one bound. Neither speaks to roll augmentation, which is a training mechanism rather than a projection — and that is exactly the half the single-seed ablation leaves unresolved.

The learned planner loses to the classical ones. Table 1 is unambiguous: RRT-Connect solves essentially every problem in 0.28 s of single-core CPU time, while the best learned arm reaches 55.8% per sample even trained to convergence. Part of the shortfall is capacity — the model is small, at 2.16M parameters — and part is that a 64-waypoint sampler from an unconditional prior is a weaker instrument than a complete planner on a problem with an exact SDF. What it is *not*, any longer, is training budget: Sec. V-H runs both arms to a plateau. We draw the comparison anyway, because a representation result measured only against other configurations of itself is not anchored to anything.

We are careful about how the underfitting claim is supported, because the obvious support is confounded. Comparing the epoch-averaged training loss against a validation loss computed from the exponential-moving-average weights is not a generalisation gap: the training figure lags, since it averages over an epoch during which the weights were still improving, and the two figures are computed from different weights, with the EMA weights typically the better ones. Both effects push validation below training regardless of fit. We therefore score a fixed slice of the training set with the same EMA weights and the same code path as validation, and read the gap only from that like-for-like pair. The direct evidence is the longer run of Sec. V-H: at fixed data and fixed capacity, three times the optimisation is worth +8.0 points of held-out collision-free rate, which is what underfitting looks like measured on the task rather than on the loss.

The claim we make is comparative and holds at every point of the scaling curve; the claim we do not make is that this planner should be used.

Scope of the negative result. The claim that the sixth degree of freedom needs no treatment is specific to this setting, and relies on the training distribution supplying implicit roll diversity through many start–goal directions. A dataset with few directions per environment, or a task whose angular structure is genuinely richer — manipulation with a gripper, or non-holonomic dynamics — could plausibly reverse it. That is precisely why we advocate measuring r rather than generalising our answer: the measurement is cheap and well defined wherever a group acts, even where our conclusion does not carry over. We do not claim it is predictive — Sec. V-E shows it is not.

The comparison we previously could not make. Corollary 5 bounds post-hoc symmetrisation of a given field; a constrained architecture searches a different hypothesis class and is not covered by it. Earlier versions of this work were careful not to borrow the bound for that purpose and recorded the experiment as untested rather than unpromising. Sec. V-F now reports it, and both halves of that caution were justified. The constrained backbone is worth +0.05 points at convergence, so the conclusion the bound would have suggested happens to hold — but it is worth roughly a fourfold saving in optimisation, which no reading of r anticipated or could have. A residual measured on a converged model says nothing about the path taken to reach it.

Frame averaging also improves endpoint error by roughly 35% ($0.0020 \to 0.0013$) on every checkpoint tested — a real effect that does not propagate to the collision-free rate, because endpoint precision was never the failure mode.

Dataset diversity. The dataset is less diverse than its nominal size suggests: 30 near-duplicate trajectories per pair means 10.8M paths carry on the order of 360k distinct queries. This does not affect the comparison, since both arms train on identical data, but it bears on how additional data should be generated — cutting trajectories per pair from 30 to 5 would buy roughly six times the environments for the same generation cost. We no longer read Fig. 2 as saying environments are the single binding constraint, since Sec. V-H shows optimisation buys a comparable amount at fixed data; the honest statement is that both are binding and neither has been pushed to saturation.

Ablations must be seed-matched, and at n=3 they are underpowered. Comparing a three-seed mean against a single-seed ablation would put roll augmentation at +4.4 points. Seed 0 happens to be the worst draw for *both* arms (31.3 and 30.1 against 36.0/35.8 and 35.9/34.8), so the mismatch manufactured most of the effect; run at matched seeds it is $+0.8 \pm 0.3$. The lesson generalises. The treatment arm's across-seed spread is 1.6 points, five times the paired spread of the ablation, so any ablation compared unpaired at this scale is measuring seed noise. Even paired, n=3 gives two degrees of freedom and a critical t of 4.3: we can bound these mechanisms as small but cannot resolve whether they are zero. All four cells of the main grid now carry three seeds; several ablations, and the converged DDPM cells, remain single seed and are reported as point estimates.

Extending beyond two points in $\mathbb{R}^3$. The reduction needs a frame derivable from the query in closed form, which is what a start and a goal *position* supply. Two extensions are worth distinguishing. In task-space planning for a manipulator — where the state is an end-effector pose — the construction carries over unchanged, and Sec. V-K is already that case: an SE(3) state with the frame built from the two endpoint positions and the orientations left to the network. In joint-space planning for a 7-DoF arm it does not carry over as written, and we want to be precise about why rather than gesture at "higher dimensions". A rigid motion of the workspace does not act on a joint vector $\mathbf{q} \in \mathbb{R}^7$ by any linear map: the symmetry is a workspace symmetry, and the forward kinematics do not intertwine it with joint coordinates. What survives is narrower and still useful — for an arm on a fixed base, a rotation of the scene about the base axis is realised exactly by an offset on the first joint, so a one-parameter canonicalisation is available in joint space, and the remaining five degrees of freedom are not. Whether the benefit measured here survives that reduction from five recoverable degrees of freedom to one is the open question, and it is a sharper question than "does it scale to 7 DoF".

Two state spaces, one workspace family. Sec. V-K replicates the pattern in a second state space — a rigid body whose state is a full pose — which rules out the reading that the result depends on a point-shaped state. It does not broaden the workspace family: both domains use the same clutter generator and the same planner architecture, and every SE(3) cell is single seed. Whether the gap survives in a manipulator configuration space, where the start–goal frame is not read off two points in $\mathbb{R}^3$, is open, and it is the setting where we would most expect the construction to need rethinking rather than reimplementing.

Reproducibility. The benchmark, the expert pipeline, both learned arms, the classical baselines, and the residual diagnostic are one codebase; every number in this paper is produced by a script that takes the checkpoint and the held-out shard range as arguments, and the figures are regenerated from those outputs rather than transcribed. The equivariance and geometry test suites described in Sec. IV-E run in seconds and are the intended entry point for anyone reimplementing the reduction, since they encode the failure modes that are silent at training time.

Future work, in priority order. *(1) Making r vary deliberately.* Sec. V-E shows that r does not forecast what a symmetry mechanism is worth, but it shows it at three points, two of which have nearly indistinguishable r. The fair objection is that nothing varied. The gauge is a deterministic function of $\hat{\mathbf{x}}$ and the data supplies many start–goal directions per environment, so restricting training to a narrow cone of directions should starve the model of implicit roll diversity and drive r up. Sweeping the cone width at fixed training-set size and fixed gradient-step count turns one measurement into a curve, and would establish either that r predicts the benefit of post-hoc symmetrisation (which Corollary 5 suggests and Sec. V-H now weakly supports) or that it does not even then, which is the harder negative. *(2) Additional seeds on the local-geometry arms*, which are single-seed where every other comparison here carries three. *(3) Convergence at* 250 *environments*, so the headline is converged as well as full-data; Sec. V-H establishes the plateau at 60.

(4) A Brownian-bridge prior. The flow begins from an isotropic Gaussian that knows nothing of the query, so it must learn both to produce a path and to place its endpoints; a bridge pinned at start and goal would leave only the obstacle-avoidance detour to be learned. We had excluded it on the grounds that such a prior is not isotropic and would break the distributional guarantee at $t = 0$. That reasoning was wrong, and we correct it here rather than repeat it: a bridge with i.i.d. coordinate noise has covariance $\sigma^2\gamma(1-\gamma)I_3$ at path fraction γ, a multiple of the identity, so $R \cdot \mathrm{Bridge}(\mathbf{s}, \mathbf{g}) \overset{d}{=} \mathrm{Bridge}(R\mathbf{s}, R\mathbf{g})$ for every R, and a roll about $\hat{\mathbf{x}}$ fixes $\mathbf{s}$, $\mathbf{g}$ and every point of the mean segment, leaving the law invariant outright. The bridge preserves exactly the SO(2) that survives the reduction. What it genuinely costs is different: the prior becomes query-dependent, so the guarantee at $t = 0$ becomes an equivariance rather than an invariance state-

ment, and the covariance is *singular* at both endpoints, so the regression target there carries no noise. The second is the real hazard, and the standard remedy is a variance floor.

VII Conclusion

The model presented here is exactly SE(3)-equivariant by construction for five of six degrees of freedom, and measured to be 98% equivariant for the sixth without any mechanism enforcing it. The benefit is concentrated in the five: the mechanisms built for the sixth are worth +0.1 and +0.8 points against +30.1, and neither is distinguishable from zero at three seeds. Within the five, recentring and axis alignment split the benefit 41/59.

The recommendation is correspondingly simple, and we believe it applies beyond this benchmark: canonicalise the degrees of freedom that admit a continuous canonical form — the query usually supplies more of them than one expects — and then *measure* the residual non-equivariance rather than assuming it. The measurement costs one forward pass.

What that measurement licenses is worth stating exactly, since it is narrower than it is tempting to write. It says that *symmetrising this model* is worth almost nothing here, because almost none of this model's error is a symmetry violation. It does not say that a network built equivariant from the start would fail, since such a network searches a different hypothesis class and could converge somewhere better. The claim is about where the error lives, not about what a different architecture could achieve. Turning r into a genuine model-selection rule would require showing that it predicts the gain from constrained architectures across tasks with differing amounts of non-equivariance, including tasks where the symmetry genuinely fails; that is future work, and we state it as such rather than implying it here.

References

[1] M. S. Albergo and E. Vanden-Eijnden. Building normalizing flows with stochastic interpolants. In *ICLR*, 2023.

[2] M. M. Bronstein, J. Bruna, T. Cohen, and P. Veličković. Geometric deep learning: Grids, groups, graphs, geodesics, and gauges. *arXiv:2104.13478*, 2021.

[3] J. Carvalho, A. T. Le, M. Baierl, D. Koert, and J. Peters. Motion planning diffusion: Learning and planning of robot motions with diffusion models. In *IROS*, 2023.

[4] C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. Burchfiel, and S. Song. Diffusion policy: Visuomotor policy learning via action diffusion. In *RSS*, 2023.

[5] T. S. Cohen and M. Welling. Group equivariant convolutional networks. In *ICML*, 2016.

[6] C. Deng, O. Litany, Y. Duan, A. Poulenard, A. Tagliasacchi, and L. Guibas. Vector neurons: A general framework for SO(3)-equivariant networks. In *ICCV*, 2021.

[7] B. Elesedy and S. Zaidi. Provably strict generalisation benefit for equivariant models. In *ICML*, 2021.

[8] A. Fishman, A. Murali, C. Eppner, B. Peele, B. Boots, and D. Fox. Motion policy networks. In *CoRL*, 2022.

[9] F. B. Fuchs, D. E. Worrall, V. Fischer, and M. Welling. SE(3)-transformers: 3D roto-translation equivariant attention networks. In *NeurIPS*, 2020.

[10] M. Geiger and T. Smidt. e3nn: Euclidean neural networks. *arXiv:2207.09453*, 2022.

[11] N. Gruver, M. Finzi, M. Goldblum, and A. G. Wilson. The Lie derivative for measuring learned equivariance. In *ICLR*, 2023.

[12] M. Janner, Y. Du, J. B. Tenenbaum, and S. Levine. Planning with diffusion for flexible behavior synthesis. In *ICML*, 2022.

[13] S.-O. Kaba, A. K. Mondal, Y. Zhang, Y. Bengio, and S. Ravanbakhsh. Equivariance with learned canonicalization functions. In *ICML*, 2023.

[14] J. J. Kuffner and S. M. LaValle. RRT-connect: An efficient approach to single-query path planning. In *ICRA*, 2000.

[15] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le. Flow matching for generative modeling. In *ICLR*, 2023.

[16] X. Liu, C. Gong, and Q. Liu. Flow straight and fast: Learning to generate and transfer data with rectified flow. In *ICLR*, 2023.

[17] J. Milnor. Analytic proofs of the "hairy ball theorem" and the Brouwer fixed point theorem. *American Mathematical Monthly*, 85(7):521–524, 1978.

[18] A. van den Oord, S. Dieleman, H. Zen, K. Simonyan, O. Vinyals, A. Graves, N. Kalchbrenner, A. Senior, and K. Kavukcuoglu. WaveNet: A generative model for raw audio. *arXiv:1609.03499*, 2016.

[19] E. Perez, F. Strub, H. de Vries, V. Dumoulin, and A. Courville. FiLM: Visual reasoning with a general conditioning layer. In *AAAI*, 2018.

[20] O. Puny, M. Atzmon, H. Ben-Hamu, I. Misra, A. Grover, E. J. Smith, and Y. Lipman. Frame averaging for invariant and equivariant network design. In *ICLR*, 2022.

[21] C. R. Qi, H. Su, K. Mo, and L. J. Guibas. PointNet: Deep learning on point sets for 3D classification and segmentation. In *CVPR*, 2017.

[22] A. H. Qureshi, A. Simeonov, M. J. Bency, and M. C. Yip. Motion planning networks. In *ICRA*, 2019.

[23] H. Ryu, H.-i. Lee, J.-H. Lee, and J. Choi. Equivariant descriptor fields: SE(3)-equivariant energy-based models for end-to-end visual robotic manipulation learning. In *ICLR*, 2023.

[24] J. Schulman, Y. Duan, J. Ho, A. Lee, I. Awwal, H. Bradlow, J. Pan, S. Patil, K. Goldberg, and P. Abbeel. Motion planning with sequential convex optimization and convex collision checking. *IJRR*, 33(9):1251–1270, 2014.

[25] A. Simeonov, Y. Du, A. Tagliasacchi, J. B. Tenenbaum, A. Rodriguez, P. Agrawal, and V. Sitzmann. Neural descriptor fields: SE(3)-equivariant object representations for manipulation. In *ICRA*, 2022.

[26] N. Thomas, T. Smidt, S. Kearnes, L. Yang, L. Li, K. Kohlhoff, and P. Riley. Tensor field networks: Rotation- and translation-equivariant neural networks for 3D point clouds. *arXiv:1802.08219*, 2018.

[27] J. Urain, N. Funk, J. Peters, and G. Chalvatzaki. SE(3)-DiffusionFields: Learning smooth cost functions for joint grasp and motion optimization through diffusion. In *ICRA*, 2023.

[28] D. Wang, S. Hart, D. Surovik, T. Kelestemur, H. Huang, H. Zhao, M. Yeatman, J. Wang, R. Walters, and R. Platt. Equivariant diffusion policy. In *CoRL*, 2024.

[29] J. Yang, Z. Cao, C. Deng, R. Antonova, S. Song, and J. Bohg. EquiBot: SIM(3)-equivariant diffusion policy for generalizable and data efficient learning. In *CoRL*, 2024.

[30] M. Zucker, N. Ratliff, A. D. Dragan, M. Pivtoraiko, M. Klingensmith, C. M. Dellin, J. A. Bagnell, and S. S. Srinivasa. CHOMP: Covariant Hamiltonian optimization for motion planning. *IJRR*, 32(9-10):1164–1193, 2013.